20251120 $\quad A, B \in M_n(\mathbb{R})$

We say that A is <u>similar</u> to B if

$\exists \underbrace{P \in GL_n(\mathbb{R})}_{P \text{ invertible}}$ s.t. $A = PBP^{-1}$

<u>Def</u> $A \in M_n(\mathbb{R})$ is <u>orthogonally</u> <u>diagonalizable</u> if

$\exists Q \in O(n)$ s.t. $A = Q \Lambda Q^T \left(= Q \Lambda Q^{-1}\right)$

where Λ is diagonal.

$O(n) = \{A \in M_n(\mathbb{R}) \mid A^{-1} = A^T\}$

set of orthogonal matrices.

<u>Thm</u> (Real Spectral thm) For $A \in M_n(\mathbb{R})$, TFAE

① $A = A^T$

orthonormal basis

② $\exists$ ONB for $\mathbb{R}^n$ s.t. it consisits of eigenvectors of A.

③ A is <u>orthogonally</u> <u>diagonalizable</u>, i.e., $\exists Q \in O(n)$ and Λ diagonal matrix s.t.

$$A = Q \Lambda Q^T$$

<u>Thm</u> Let $A \in M_n(\mathbb{R})$, $A = A^T$, then all the eigen values of A are real.

<u>Pf</u> Let $\lambda \in \mathbb{C}$ be an eigenval. of A s.t. $Av = \lambda v$

$v \in \mathbb{R}^n \setminus \{0\}$. Note that $\overline{A}^T = \overline{A^T} = A$, then

$$\lambda \|v\|^2 = \lambda \overline{v}^T v = \overline{v}^T(\lambda v) = \overline{v}^T A v = \overline{v}^T \overline{A}^T v$$

$$= \overline{(Av)}^T v = \overline{\lambda}\, \overline{v}^T v = \overline{\lambda}\, \|v\|^2$$

$$\Rightarrow (\lambda - \overline{\lambda}) \underbrace{\|v\|^2}_{\neq 0 \text{ b/c } v \neq 0} = 0$$

$$\Rightarrow \lambda = \overline{\lambda} \Rightarrow \lambda \in \mathbb{R} \qquad \square$$

In $\mathbb{C}^n$, the standard inner product is given by

$$\langle x, y \rangle := \sum_{i=1}^{n} \overline{x_i}^T y_i \qquad \forall x, y \in \mathbb{C}^n$$

$$\Downarrow$$

$$\left(\text{s.t.} \quad \underset{\|x\|^2}{\overset{\shortparallel}{\langle x, x \rangle}} \geq 0 \quad \forall x \right) \qquad \begin{aligned} \langle x, \lambda y \rangle &= \lambda \langle x, y \rangle \\ \langle \lambda x, y \rangle &= \overline{\lambda} \langle x, y \rangle \end{aligned}$$

__Thm__ $A \in M_n(\mathbb{R})$, $A^T = A$, if λ & μ are distinct eigenvalues of A w/ e-vec v & w, i.e.,

$$Av = \lambda v \quad \& \quad Aw = \mu w$$

then $v \perp w$, i.e., $\langle v, w \rangle = v^T w = 0 \in \mathbb{R}$

__Pf__ $\mu \langle v, w \rangle = \langle v, \mu w \rangle = \langle v, Aw \rangle$

$$= \langle Av, w \rangle = \langle \lambda v, w \rangle = \overline{\lambda} \langle v, w \rangle$$

$$= \lambda \langle v, w \rangle$$

$$\Rightarrow \underbrace{(\mu - \lambda)}_{\neq 0} \langle v, w \rangle = 0 \Rightarrow \langle v, w \rangle = 0 \qquad \square$$

ex $\quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$$\det(A - \lambda I_3) = \det\begin{pmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{pmatrix} = 0$$

$$\Rightarrow \lambda = 0, 0, 3$$

In fact, $\chi_A(A) = \det(A - \lambda I_3)$

$$= -\lambda^2(\lambda - 3)$$

$$= -(\lambda - 0)^2(\lambda - 3)^1$$

we say that

$\lambda = 0$ has algebraic multiplicity of 2

& $\lambda = 3$ has ________________ of 1

If $\lambda = 0$, then solve $((A - \lambda I)v = 0)$

$$\begin{bmatrix} 1-0 & 1 & 1 \\ 1 & 1-0 & 1 \\ 1 & 1 & 1-0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c_1\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_2\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

we say that the eigen space of $\lambda = 0$ is

$$V_{\lambda=0} = \ker(A - 0I) = \operatorname{span}\left\{ \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}_{= v_1}, \underbrace{\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}}_{= v_2} \right\}$$

so for $\lambda = 0$, we can find 2 linearly independent eigenvectors, i.e., $\dim V_{\lambda=0} = 2$, we say that the geometric multiplicity of $\lambda = 0$ is 2

FACT 1 For a fixed eigenvalue λ

geo. mult. of $\lambda \leq$ alg. mult. of λ

FACT 2 a matrix is diagonalizable $\iff$ For each eigenvalue λ,

geo. mult. of $\lambda =$ alg. mult. of λ

For $\lambda = 3$, solve

$$\begin{bmatrix} 1-3 & 1 & 1 \\ 1 & 1-3 & 1 \\ 1 & 1 & 1-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c \underset{\overset{\shortparallel}{v_3}}{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}, \quad c \in \mathbb{R}$$

$$\Rightarrow V_{\lambda=3} = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

geo. mult. of $(\lambda=3) =$ alg. mult. of $(\lambda = 3) = 1$

$$\Rightarrow A \underbrace{\begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}}_{P} = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \underbrace{\begin{bmatrix} \boxed{\begin{matrix} 0 & \\ & 0 \end{matrix}} & \\ & \boxed{3} \end{bmatrix}}_{\Lambda}$$

Although $v_1 \perp v_3$ & $v_2 \perp v_3$ b/c $0 \neq 3$

but $v_1 \not\perp v_2$, so $P \notin O(n)$

Let $q_3 = \dfrac{v_3}{\|v_3\|} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} / \sqrt{3}$

For $\lambda = 0$, perform Gram-Schmidt to $\{v_1, v_2\}$

Let $$q_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} / \sqrt{2}$$

$$q_2 = \frac{v_2 - q_1 \langle q_1, v_2 \rangle}{\| \quad '' \quad \|} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} / \sqrt{6}$$

$$\Rightarrow Q = \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix} \in O(3)$$

$$\Rightarrow A = \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} 0 & & \\ & 0 & \\ & & 3 \end{bmatrix} \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix}^T$$

$$\Rightarrow A = \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 0 & & \\ & 0 & \\ & & 3 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & +1/\sqrt{2} & 0 \\ -1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

In general, for any $A \in M_n(\mathbb{C})$, we can decompose A as $$A = PJP^{-1}, \quad P \in GL_n(\mathbb{R})$$

and J is <u>block-diagonal</u>

$$J = \begin{bmatrix} \boxed{J_1} & & & \\ & \boxed{J_2} & & \\ & & \ddots & \\ & & & \boxed{J_m} \end{bmatrix}$$

The square matrices J_i's are called Jordan blocks s.t.

$$J_i = \begin{bmatrix} \lambda_i & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_i \end{bmatrix}_{m_i \times m_i} = \lambda_i I + \overset{\text{nilpotent}}{\overset{\downarrow}{N}}, \quad \text{w/ } N = \begin{bmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{bmatrix}$$

This decomposition is called Jordan decomposition.

J is called Jordan Normal/Canonical Form

(JNF/JCF)

e.g. $$\exp(tJ) = \begin{bmatrix} \boxed{e^{tJ_1}} & & 0 \\ & \ddots & \\ 0 & & \boxed{e^{tJ_m}} \end{bmatrix}$$

w/

$$e^{tJ_i} = e^{(\lambda_i I + N)t} \overset{\lambda_i I \,\&\, N \text{ commutative}}{\overset{\downarrow}{=}} e^{\lambda_i I t} e^{tN}$$

$$= e^{\lambda_i t}\left[I + tN + \frac{(tN)^2}{2!} + \cdots \frac{(tN)^{m_i-1}}{(m_i-1)!} \right]$$

$$= e^{\lambda_i t} \begin{bmatrix} 1 & t & t^2/2 & \cdots & \frac{t^{m_i-1}}{(m_i-1)!} \\ & \ddots & t & \ddots & \vdots \\ & & \ddots & \ddots & t^2/2 \\ & & & \ddots & t \\ 0 & & & & 1 \end{bmatrix}$$

$$\Rightarrow e^{tA} = P e^{tJ} P^{-1}$$

<u>FACT</u> JNF unique up to reordering the Jordan block.

(Can uniquely decompose A as $A = S + \eta$
s.t. S is diagonalizable & η is nilpotent)

- For 2×2 matrices w/ e-value λ

$$\begin{bmatrix} \boxed{\lambda} & \\ & \boxed{\lambda} \end{bmatrix} \qquad \begin{bmatrix} \lambda & 1 \\ & \lambda \end{bmatrix}$$

$2 = 1 + 1$ $\qquad$ $2 = 2$

- For 3×3 matrices w/ e-val λ

$$\begin{bmatrix} \boxed{\lambda} & & \\ & \boxed{\lambda} & \\ & & \boxed{\lambda} \end{bmatrix}$$

$3 = 1 + 1 + 1$

alg. mult = 3

geo mult = 3

$$\begin{bmatrix} \boxed{\lambda} & & \\ & \lambda & 1 \\ & & \lambda \end{bmatrix}$$

$3 = 2 + 1$

alg. mult = 3

geo. mult = 2

$$\begin{bmatrix} \lambda & 1 & 0 \\ & \lambda & 1 \\ & & \lambda \end{bmatrix}$$

$3 = 3$

alg. mult = 3

geo. mult = 1

- For 4×4 matrices w/ e-val λ

$$\begin{bmatrix} \boxed{\lambda} & & & \\ & \boxed{\lambda} & & \\ & & \boxed{\lambda} & \\ & & & \boxed{\lambda} \end{bmatrix}$$

$4 = 1 + 1 + 1 + 1$

alg. mult = 4

geo mult = 4

$$\begin{bmatrix} \lambda & 1 & & \\ & \lambda & & \\ & & \lambda & 1 \\ & & & \lambda \end{bmatrix}$$

$4 = 2 + 2$

alg. mult = 4

geo mult = 2

$$\begin{bmatrix} \boxed{\lambda} & & & \\ & \lambda & 1 & \\ & & \lambda & 1 \\ & & & \lambda \end{bmatrix}$$

$4 = 3 + 1$

alg. mult = 4

geo mult = 2

$$\begin{bmatrix} \boxed{\lambda} & & & \\ & \boxed{\lambda} & & \\ & & \lambda & 1 \\ & & & \lambda \end{bmatrix}$$

$4 = 2 + 1 + 1$

alg. mult = 4

geo mult = 3

$$\begin{bmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \lambda & 1 \\ & & & \lambda \end{bmatrix}$$

$4 = 4$

alg. mult = 4

geo mult = 1